

MAHMOUD ZAKI

Cairo, Egypt | +20 111 533 9933 / +20 100 258 0455 | M.zaki.khetabi@gmail.com | LinkedIn Profile

PROFESSIONAL SUMMARY

Results-driven and highly technical Senior Production Shift Leader with over 20 years of robust experience in cement manufacturing, process optimization, and large-scale plant operations. Proven expertise in managing complex FLS and KHD cement production lines, maximizing equipment reliability, and spearheading complex commissioning projects. Exceptionally skilled in multi-fuel systems, with a demonstrated ability to optimize combustion matrixes and seamlessly transition between conventional and alternative fuels (AFR, Petcoke, Mazut, and Hydrogen) to achieve massive energy savings and emission reductions. A decisive leader with a strong scientific background, expert in industrial diagnostics, root-cause troubleshooting, and fostering a strict zero-accident HSE culture.

KEY SKILLS

- **Industrial & Technical Leadership:** Production Shift Leadership, Pyro-Processing Optimization, Multi-Fuel Grinding & Combustion Management, High-Performing Team Management.
- **System Specializations:** FLS & KHD Cement Technologies, Preheater Rotary Kilns, SF Clinker Coolers, Vertical Raw Mills (Atox 42.5 / 40), Ball Mills.
- **Commissioning & Project Execution:** Hot Disc Systems, Solid Fuel (Coal/Petcoke) Mill Systems, Plant Line Start-ups & Upgrades.
- **Process Diagnostics & Quality:** Industrial Troubleshooting, Root-Cause Analysis (RCA), Cement Chemistry Optimization, XRF Analysis, EN 197-1 Standards.
- **Safety Excellence:** Corporate HSE Compliance, Hazard Identification, Zero Lost Time Injuries (LTI) Frameworks.

PROFESSIONAL EXPERIENCE

RHI - Ramliya Project

Senior Production Shift Leader | Feb 2021 - Present

- Direct full-scale integrated plant operations from the Central Control Room (CCR), ensuring consistent achievement of daily clinker and cement volumetric production targets.
- Spearhead real-time process diagnostics and technical troubleshooting models to resolve clinkerization and pyro-processing bottlenecks, maximizing line uptime.
- Orchestrate and supervise major annual shutdown activities, managing multi-disciplinary teams across mechanical, electrical, and refractory overhauls safely.
- Optimize multi-fuel combustion operations, managing transitions between alternative fuels (AFR & Hydrogen) and

traditional sources to lower heat consumption and reduce carbon footprint.

- Train, mentor, and supervise shift technical staff to improve operational efficiency, competency, and absolute adherence to strict safety guidelines.

ASEC - Al Minya Project

Production Shift Leader | May 2020 - Feb 2021

- Managed comprehensive plant shift production loops, leading control room operators and site technicians to maintain process stability and baseline target parameters.
- Coordinated cross-functional maintenance schedules with electrical and mechanical departments to execute prompt corrective interventions during process deviations.
- Enforced corporate health and safety procedures across plant sections, successfully sustaining zero-incident operational cycles.

ASEC - BMIC Project

Production Shift Leader | Apr 2019 - May 2020

- Supervised integrated plant operations covering kilns, raw mills, cement mills, and solid fuel preparation networks from the control station.
- Implemented strategic adjustments to counter raw material variations, ensuring stable chemical parameter control and high clinker quality factors.
- Analyzed process trends to detect abnormalities early, reducing emergency mechanical stops and maximizing line thermal efficiency.

FLSmidth - Ramliya Project

Production Shift Leader | Jul 2017 - Apr 2019

- Led daily shifting operations for FLSmidth process management, aligning workflow streams and optimizing equipment operational integrity factors.
- Collaborated closely with safety and maintenance departments to streamline preventative maintenance schedules, minimizing process related downtime.
- Monitored alternative fuel feeding rates, ensuring balanced combustion chemistry within the kiln preheater columns.

FLSmidth - Ramliya Project

Line Operator (Kiln & Raw Mill) | Jul 2012 - Jul 2017

- Operated and dynamically stabilized a high-capacity preheater rotary kiln (5x80m) and SF 4x6F grate cooler system yielding 6,000 TPD.
- Regulated vertical raw mill (Atox 42.5) operational variables, maximizing material throughput while preserving exact product fineness parameters.
- Managed multi-fuel combustion operations (AFR, coal, petcoke, natural gas, diesel) through the specialized Hot Disc system to maintain an optimal burning profile.

ASEC - Sinai Gray Cement Project

Kiln Operator | Oct 2009 - Jul 2012

- Actively participated in the hot commissioning, testing, and initial startup phases of Line 2 (5,000 TPD),

establishing long-term kiln thermal stability.

- Monitored pyro-processing parameters, managing clinker liter weights and free lime levels within strict quality targets.

ASEC - Sinai Gray Cement Project

Mills Operator | Dec 2008 - Oct 2009

- Controlled an Atox 40 raw mill and twin cement ball mills, optimizing grinding circuit configurations to maximize hourly tonnage and grinding stability.

ASEC - Sinai Gray Cement Project

Q.C. Chemist | Oct 2006 - Dec 2008

- Conducted rigorous chemical and elemental analyses using XRF technology from raw mix to final product, validating EN 197-1 standard compliance.
- Evaluated daily laboratory calibration benchmarks and managed precise chemical raw mix proportioning equations.

SGS Egypt - Cement & Silica Sand Division

Chemist | Aug 2005 - Oct 2006

- Performed analytical evaluation and technical testing routines for cement and silica sand output compliance against Egyptian and European industrial standards.

CERTIFICATIONS & TRAINING

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OSHA Industrial Safety Course – The American University in Cairo (AUC)
- **Comprehensive Cement Manufacturing Technology Course** – ASEC Academy

EDUCATION

Bachelor of Science (B.Sc.) in Biology

Faculty of Science, Al-Azhar University Cairo | 2000 - 2004

LANGUAGES

Arabic: Native

English: Advanced (Professional working proficiency)